Design, Configuration and Simulation of a Hybrid Wired–Wireless Network

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1. Introduction

The growing need to use networked communication in organisational settings has led to the significance of scalable and reliable network topology. Modern networks need to combine wired and wireless access technologies to accommodate a broad range of devices and user behaviours providing mobility and guaranteeing its secure and systematic connection. This can be done by a thorough planning of IP addressing, switching technology-based segmentation, secure wireless deployment, router configuration, and routing protocols to ensure a regular visibility between subnets.

This report is a report of the entire design, configuration and testing of a hybrid network that was implemented on Cisco Packet Tracer. The development goal was to have a complete-fledged communication system that will support secure wireless client access, static addressing of server-centric hosts, and stable connection between different segments of the network that are spread across two routers. It is based on the hierarchical principles of networking, which guarantees modular scalability and a very clear control of the traffic in the environment. The undertaken work validates the importance of the logical separation, systematically addressed addressing, and stricter verification in order to reach a successful and mutually connected network serving mobile and fixed devices.

2. Network Design Overview

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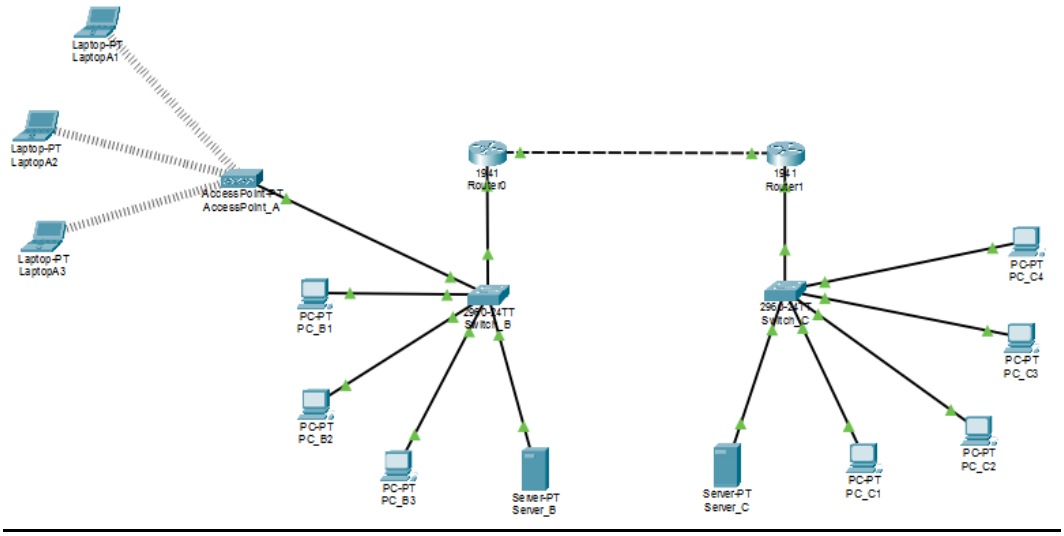

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Building configuration...
[OK]
Router0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#router rip
Router0(config-router)# version 2
Router0(config-router)# no auto-summary
Router0(config-router)# network 191.50.44.0
Router0(config-router)# network 10.10.10.0
Router0(config-router)#exit
Router0(config)#end
Router0#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router0#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router0#
```

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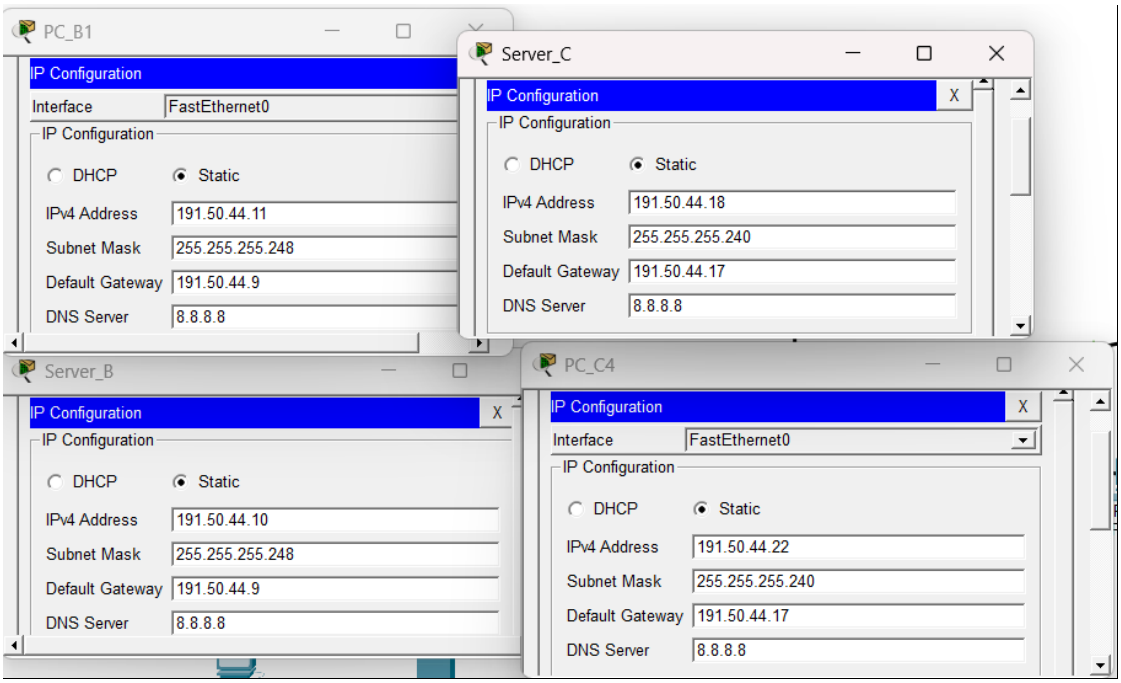

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Switch_B

IOS Command Line Interface

```
Switch(config)#! Trunk port toward Router0 (G0/0 subinterfaces)
Switch(config)#interface FastEthernet0/1
Switch(config-if)# switchport trunk encapsulation dot1q
                                  ^
% Invalid input detected at '^' marker.

Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)#
Switch(config)#! Access port to AccessPoint (VLAN 10)
Switch(config)#interface FastEthernet0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
Switch(config-if)# exit
Switch(config)#
Switch(config)#! Access ports for Wired hosts (VLAN 20) - adjust range for number of PCs
Switch(config)#interface range FastEthernet0/3 - 6
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit
Switch(config)#
Switch(config)#end
```

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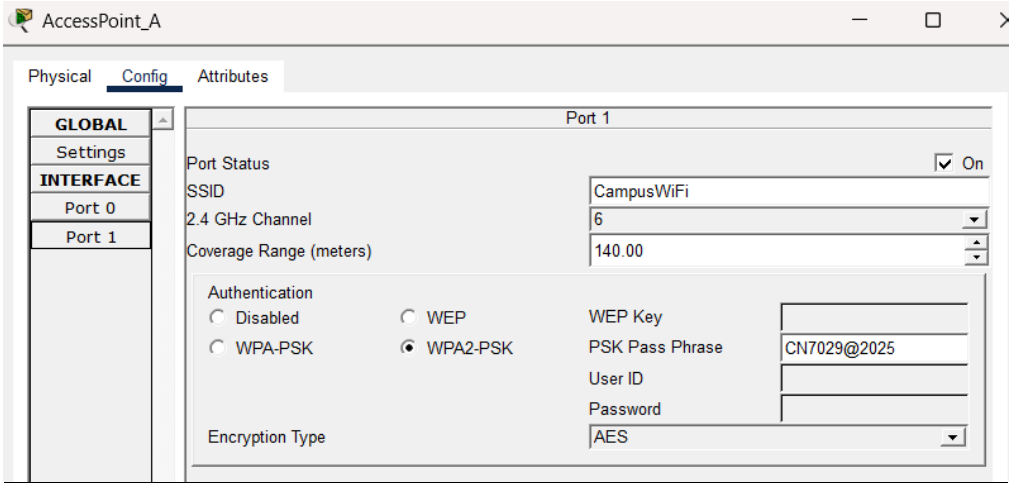

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Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#router rip
Router1(config-router)# version 2
Router1(config-router)# no auto-summary
Router1(config-router)# network 191.50.44.0
Router1(config-router)# network 10.10.10.0
Router1(config-router)#exit
Router1(config)#end
Router1#write memory
%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
[OK]
Router1#ping 10.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

[redacted]

[redacted]

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router0#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router0(config)#
Router0(config)#
Router0(config)#interface gigabitEthernet0/0
Router0(config-if)# no ip address
Router0(config-if)# no shutdown
Router0(config-if)#exit
Router0(config)#
Router0(config)#! Router-to-router link on G0/1
Router0(config)#interface gigabitEthernet0/1
Router0(config-if)# ip address 10.10.10.1 255.255.255.252
Router0(config-if)# no shutdown

Router0(config-if)#exit
Router0(config)#
Router0(config)#end
Router0#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Building configuration...
[OK]
```

[redacted]

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router1#
%SYS-5-CONFIG_I: Configured from console by console

Router1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     191.50.44.17    YES manual up                    up
GigabitEthernet0/1     191.50.44.29    YES manual administratively down down
Vlan1                  unassigned      YES NVRAM  administratively down down
Router1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router1(config)#
Router1(config)#interface gigabitEthernet0/1
Router1(config-if)# ip address 10.10.10.2 255.255.255.252
Router1(config-if)# no shutdown

Router1(config-if)#exit
Router1(config)#
Router1(config)#end
Router1#write memory
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Building configuration...
```

[redacted]

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[redacted]

[redacted]

Router0

IOS Command Line Interface

```
Router0>enable
Router0#show ip dhcp pool

Pool WirelessPool :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 6
 Leased addresses               : 3
 Excluded addresses             : 1
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 191.50.44.1          191.50.44.1      - 191.50.44.6       3    / 1 |   / 6
Router0#show run | section dhcp
ip dhcp excluded-address 191.50.44.1
ip dhcp pool WirelessPool
 network 191.50.44.0 255.255.255.248
 default-router 191.50.44.1
 dns-server 8.8.8.8
Router0#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
191.50.44.2      00D0.585E.69B6           --                     Automatic
191.50.44.3      0001.64DB.9100           --                     Automatic
191.50.44.4      0060.2FA9.B674           --                     Automatic
Router0#
```

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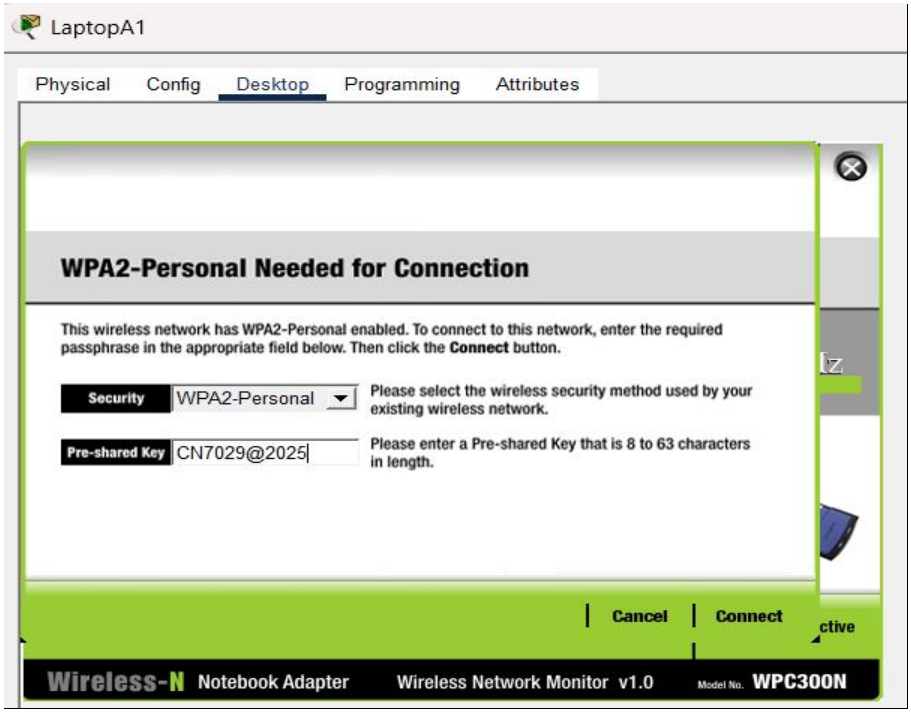

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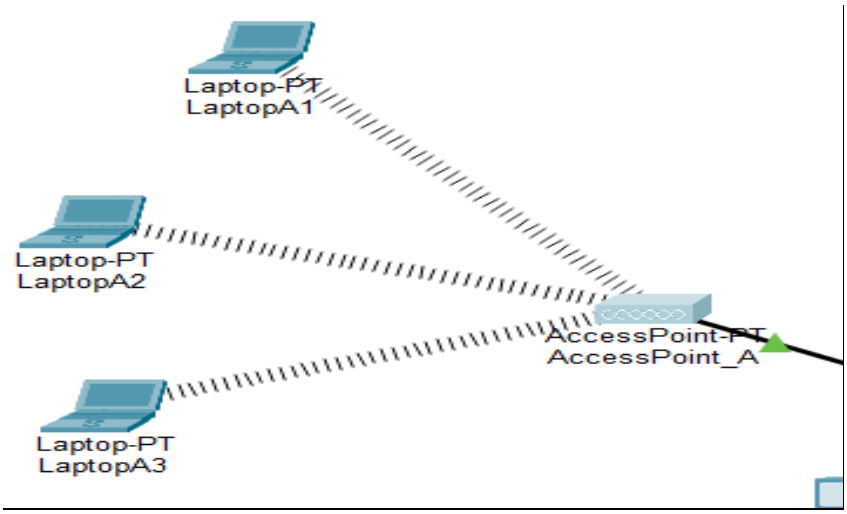

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PC_B1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 191.50.44.3

Pinging 191.50.44.3 with 32 bytes of data:

Reply from 191.50.44.3: bytes=32 time=10ms TTL=127
Reply from 191.50.44.3: bytes=32 time=10ms TTL=127
Reply from 191.50.44.3: bytes=32 time=2ms TTL=127
Reply from 191.50.44.3: bytes=32 time=5ms TTL=127

Ping statistics for 191.50.44.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 10ms, Average = 6ms
```

[redacted]

PC_B1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>ping 191.50.44.18

Pinging 191.50.44.18 with 32 bytes of data:

Request timed out.
Reply from 191.50.44.18: bytes=32 time=1ms TTL=126
Reply from 191.50.44.18: bytes=32 time=10ms TTL=126
Reply from 191.50.44.18: bytes=32 time=2ms TTL=126

Ping statistics for 191.50.44.18:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 4ms

C:\>ping 191.50.44.18

Pinging 191.50.44.18 with 32 bytes of data:

Reply from 191.50.44.18: bytes=32 time<1ms TTL=126
Reply from 191.50.44.18: bytes=32 time<1ms TTL=126
Reply from 191.50.44.18: bytes=32 time<1ms TTL=126
Reply from 191.50.44.18: bytes=32 time=4ms TTL=126

Ping statistics for 191.50.44.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

[redacted]

[redacted]

LaptopA3

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 191.50.44.10

Pinging 191.50.44.10 with 32 bytes of data:

Request timed out.
Reply from 191.50.44.10: bytes=32 time=19ms TTL=127
Reply from 191.50.44.10: bytes=32 time=8ms TTL=127
Reply from 191.50.44.10: bytes=32 time=13ms TTL=127

Ping statistics for 191.50.44.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 8ms, Maximum = 19ms, Average = 13ms

C:\>ping 191.50.44.10

Pinging 191.50.44.10 with 32 bytes of data:

Reply from 191.50.44.10: bytes=32 time=18ms TTL=127
Reply from 191.50.44.10: bytes=32 time=20ms TTL=127
Reply from 191.50.44.10: bytes=32 time=21ms TTL=127
Reply from 191.50.44.10: bytes=32 time=19ms TTL=127

Ping statistics for 191.50.44.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 18ms, Maximum = 21ms, Average = 19ms
```

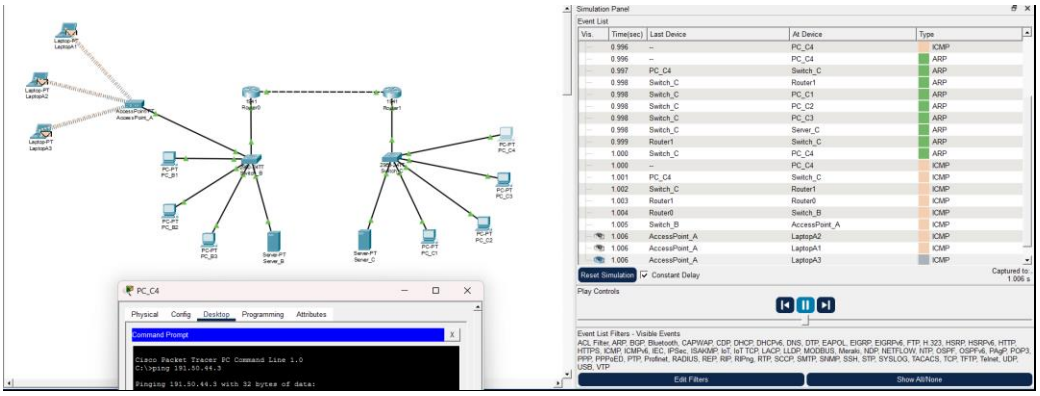
[redacted]

Server_C

Command Prompt

```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 191.50.44.3

Pinging 191.50.44.3 with 32 bytes of data:

Reply from 191.50.44.3: bytes=32 time=26ms TTL=126
Reply from 191.50.44.3: bytes=32 time<1ms TTL=126
Reply from 191.50.44.3: bytes=32 time=1ms TTL=126
Reply from 191.50.44.3: bytes=32 time=18ms TTL=126

Ping statistics for 191.50.44.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 26ms, Average = 11ms

C:\>
```

[redacted]



[redacted]

```
Router0>enable
Router0#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router0#
```

[redacted]

```
Router1>enable
Router1#ping 10.10.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router1#
```

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